

- (c) What do you mean by Alpha Beta pruning in AI? Explain in detail.
6. (a) Explain any five A. I. agents in detail.
- (b) Give the difference between Breadth First Search (BFS) and Depth First Search (DFS).
- (c) Discuss the difference between forward chaining and backward chaining.

No. of Printed Pages : 04

Following Paper ID and Roll No. to be filled in your Answer Book.

PAPER ID : 43202	Roll									
	No.	1	2	4	0	4	3	9	3	9

B. Tech. Examination, 2024-25

(Odd Semester)

BASICS OF ARTIFICIAL INTELLIGENCE

Time : Three Hours]

[Maximum Marks : 60

Note :- Attempt all questions.

SECTION-A

1. Attempt all parts of the following : 8×1=8
- (a) What do you understand by term "Intelligence"?
- (b) Write any two drawbacks of artificial intelligence.
- (c) How the soft computing is related to artificial intelligence?
- (d) Write the types of artificial intelligence.

[P. T. O.]

- (e) List any two real life examples of artificial intelligence.
- (f) Explain the term algorithm and its basic properties.
- (g) What is speech recognition in the field of artificial intelligence?
- (h) Define the term PEAS about agent.

SECTION – B

2. Attempt any two parts of the following : $2 \times 6 = 12$
- (a) Explain fuzzy set and its important operations.
 - (b) What is 8 Queen problem in artificial intelligence? Explain.
 - (c) Write in detail about AO* algorithm with example.
 - (d) Explain the difference between informed search and uninformed search.

SECTION – C

Note :- Attempt all questions. Attempt any two parts from each questions. $8 \times 5 = 40$

- 3. (a) Discuss the historical evolution of artificial intelligence.
- (b) Explain hill climbing search algorithm.
- (c) Explain in detail about first order logic.
- 4. (a) Define soft computing in detail and also write the difference between hard and soft computing.
- (b) Explain in detail about depth-limited search algorithm with example.
- (c) Discuss about Bayesian networks in detail.
- 5. (a) Explain following artificial intelligence terminologies :
 - (i) Environment
 - (ii) Pruning
 - (iii) Blackboard
 - (iv) Agents
- (b) Discuss in detail about impact of A.I. on human life.

[P. T. O.]